

Abstract

Automated classification of dermoscopic skin lesion images presents a significant challenge in computational dermatology, primarily due to severe class imbalance and high inter-class visual similarity. This study investigates and compares multiple deep learning approaches for seven-class skin lesion classification on the HAM10000 dataset, which comprises 10,015 dermoscopic images spanning seven diagnostic categories including melanoma, melanocytic nevi, and several rarer lesion types. We designed and evaluated two custom Convolutional Neural Network (CNN) architectures — a baseline model and an advanced variant incorporating batch normalization, dropout regularization, and class-weighted cross-entropy loss — alongside three ResNet18 transfer learning experiments employing different training strategies: no augmentation (Experiment A), data augmentation (Experiment B), and combined data augmentation with class weighting (Experiment C). Experimental results demonstrate that transfer learning substantially outperforms custom CNN architectures across all evaluation metrics. Among the ResNet18 variants, Experiment B achieved the best generalization balance, reaching a test accuracy of 78.95% and a macro-F1 of 0.6670, while data augmentation effectively narrowed the training–validation performance gap. Combining class weighting with augmentation destabilized training and degraded overall performance despite selectively improving minority-class recall. Confusion matrix and Grad-CAM analyses further reveal persistent MEL–NV and BKL–MEL misclassifications, reflecting the inherent visual ambiguity between benign and malignant lesions. These findings highlight the critical role of architecture selection and training strategy design in imbalanced medical image classification.

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1. Introduction and Motivation

Skin cancer is among the most prevalent malignancies globally, with melanoma representing its most lethal form due to its capacity for rapid metastasis when diagnosis is delayed. Dermatoscopy, a non-invasive imaging technique that magnifies and illuminates subsurface skin structures, has substantially improved clinical diagnostic accuracy compared to unaided visual inspection [1]. However, accurate interpretation of dermoscopic images remains highly skill-dependent and subject to inter-observer variability, motivating the development of automated computer-aided diagnosis systems. The rapid advancement of deep learning has opened new possibilities in this domain. Machine learning methods have been proposed to automate lesion analysis, improving clinical reliability and assisting physicians in diagnostic decision-making [2]. Extending these successes to fine-grained multi-class classification across the full spectrum of pigmented lesions, however, remains considerably more challenging.

The HAM10000 dataset, comprising 10,015 dermoscopic images across seven diagnostic categories, provides a widely adopted benchmark for this task, yet it introduces two fundamental obstacles that are representative of real-world clinical data. First, severe class imbalance — with melanocytic nevi accounting for approximately 67% of all samples [1] — creates a strong majority-class bias that inflates global accuracy while masking poor sensitivity on clinically critical minority classes such as melanoma and actinic keratoses. Second, high inter-class visual similarity, particularly between melanoma, benign nevi, and benign keratosis-like lesions, challenges even deep feature extractors. This study therefore systematically investigates how architectural design and training strategy — including custom CNN construction, pretrained ResNet18 fine-tuning, training-time data augmentation, and class-weighted loss formulations — jointly affect classification performance across all seven categories. Through rigorous evaluation using accuracy, macro-F1, per-class metrics, and confusion matrix analysis, we aim to provide actionable insights into effective strategies for imbalanced dermoscopic image classification.

2. Dataset Description and Preprocessing

2.1 Dataset Source

All experiments were performed on the HAM10000 dataset, which is the official training set of the ISIC 2018 Task 3: Lesion Diagnosis challenge. The dataset is publicly available through the International Skin Imaging Collaboration (ISIC) Archive and can be accessed via the following sources: ISIC 2018 Challenge page: <https://challenge.isic-archive.com/landing/2018/47/>

In this project, HAM10000 was used as the training and validation pool. For final evaluation we used the official ISIC 2018 Task 3 test set, which contains 1511 dermoscopic images with ground-truth labels (not part of the HAM10000 collection).

The train and test datasets both contain dermoscopic images from seven diagnostic categories:

Table 1. Classes Name

Label	Full name
akiec	Actinic keratosis / Bowen's disease
bcc	Basal cell carcinoma
bkl	Benign keratosis-like lesion
df	Dermatofibroma
mel	Melanoma
nv	Melanocytic nevus
vasc	Vascular lesion

2.2 Completeness Check

The HAM10000 metadata file contains 10015 image records, and all 10015 corresponding JPG images were found locally. The HAM10000 images are stored in two folders: HAM10000_images_part_1 contains 5000 images, and

HAM10000_images_part_2 contains 5015 images.

The Official ISIC 2018 Task 3 TestSet metadata contains 1512 samples. One image file, ISIC_0035068.jpg, was not present in the downloaded test image folder. This row was removed from the usable test CSV. After this correction, the final test split contains 1511 usable images.

This completeness check is important because every row used by the data loader must point to an existing image file. Removing the missing row prevents runtime loading errors and keeps the test CSV internally consistent.

2.3 Train/Validation/Test Split

The HAM10000 pool was divided into training and validation sets using a stratified split with random seed 42. The test set was not used when creating this split.

Table 2. Data set division

Split	Source	Count	Use
Train	HAM10000	8513	Training data
Validation	HAM10000	1502	Validation data
Test	Official ISIC 2018 Task 3 TestSet	1511	Held-out test data

- data/splits/train_seed42.csv
- data/splits/val_seed42.csv
- data/splits/test_school.csv

2.4 Class Distribution and Class Imbalance

The dataset is highly imbalanced. The majority class is nv, while df, vasc, and akiec have much fewer samples.

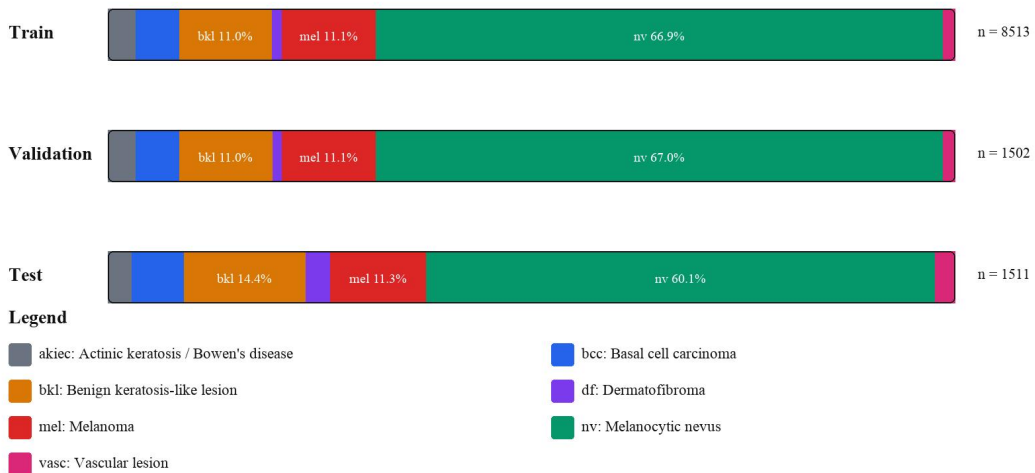
Table 3. Class distribution

Label	Full name	Train	Validation	Test
akiec	Actinic keratosis /Bowen's disease	278	49	43

bcc	Basalcell carcinoma	437	77	93
bkl	Benign keratosis-like lesion	934	165	217
df	Dermatofibroma	98	17	44
mel	Melanoma	946	167	171
nv	Melanocytic nevus	5699	1006	908
vasc	Vascular lesion	121	21	35

Class Proportion by Split

HAM10000 train/validation split and school-provided held-out test set



Note: nv is the majority class in all splits; minority classes such as df, vasc, and akiec remain underrepresented.

Figure 1. Class proportion by split.

This imbalance should be considered when interpreting later results. The exploratory analysis also shows why overall accuracy alone can be misleading for this task. Because nv accounts for the largest proportion of the training set, a classifier may obtain a high aggregate score while still performing poorly on underrepresented but clinically important categories such as mel, bcc, akiec, vasc, and df.

To support imbalance-aware training decisions, class weights were computed only from the training split using inverse class frequency: $w_c = \frac{N}{K \cdot n_c}$.

Table 4. Class weight

Label	Training-set class weight
akiec	4.375
bcc	2.783
bkl	1.302
df	12.410
mel	1.286
nv	0.213
vasc	10.051

2.5 Image Preprocessing

The original JPG images should be used instead of the low-resolution `hmnist_*` CSV files, because the original images preserve more visual detail. The exact input size can differ across experiments, but the split files and label order should remain fixed.

As a supplementary CNN-side preprocessing plan, two image preparation variants were defined. The baseline preprocessing pipeline resizes images to 64 x 64 pixels and scales pixel values to the range [0, 1]. No random augmentation, resampling, or class balancing is applied in this baseline setting, so it can serve as a simple control pipeline.

The advanced CNN preprocessing pipeline increases the image resolution to 128 x 128 pixels to preserve more lesion boundary, color transition, and texture information. Training-time augmentation is then applied through random horizontal flipping with probability 0.5 and random rotation up to 45 degrees. These transformations increase image variation during training while preserving the lesion label. After tensor conversion, images are standardized using channel means and standard deviations computed from the training split. The computed training-set mean is approximately

[0.7630, 0.5455, 0.5703], and the computed standard deviation is approximately [0.1405, 0.1519, 0.1690].

For validation and test images, preprocessing should be deterministic. It should use the same resize and normalization setting selected for the corresponding experiment, but random augmentation should not be applied. This keeps validation and test measurements fixed and reproducible.

For the ResNet preprocessing pipeline, images are resized to 224 x 224 pixels to match common pretrained input conventions. When ImageNet-pretrained weights are used, ImageNet normalization is applied using mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. The ResNet training transform includes random horizontal flip with probability 0.5, random rotation up to 20 degrees, and mild color jitter with brightness, contrast, and saturation set to 0.1. The ResNet validation and test transform uses only resizing, tensor conversion, and ImageNet normalization. The train/validation/test split files remain unchanged for both CNN and ResNet experiments.

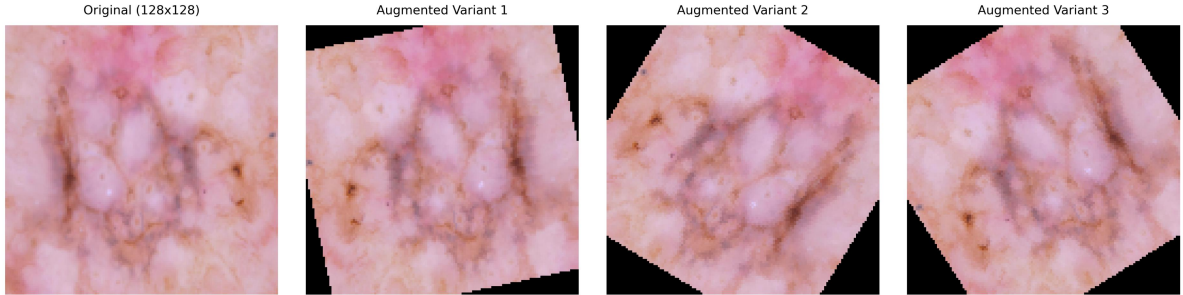


Figure 2. Example stochastic augmentation variants for a dermoscopic image.

3. Methodology

3.1 Simple Convolutional Neural Network (CNN)

To systematically isolate the effects of regularization, spatial resolution, and weighted objective functions, we designed and implemented two distinct customized Convolutional Neural Network (CNN) architectures using PyTorch.

3.1.1 Baseline Custom CNN Architecture

The baseline model serves as our experimental control, utilizing a classic, unregularized feedforward convolutional pipeline:

- **Feature Extractor:** Three consecutive convolution blocks. Each block consists of a 3×3 Convolutional layer (nn.Conv2d), followed by a Rectified Linear Unit activation (nn.ReLU) and a Max-Pooling layer (nn.MaxPool2d) for spatial downsampling.
- **Classifier:** A single fully connected linear layer mapping flattened feature maps directly to the 7-dimensional output logits.
- **Loss Function:** Standard unweighted Cross-Entropy Loss:

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{z_{i,y_i}}}{\sum_j e^{z_{i,j}}}$$

This loss function treats all misclassifications equally, driving the model to prioritize the majority class (NV) to maximize global accuracy at the expense of rare classes.

3.1.2 Advanced CNN Architecture

The Advanced CNN introduces targeted structural and mathematical modifications designed to stabilize training, prevent overfitting, and address the extreme class imbalance:

- **Batch Normalization:** Integrated immediately after each convolutional layer to stabilize internal covariate shift, smooth the optimization landscape, and facilitate faster gradient convergence.
- **Regularization (Dropout):** A dropout layer with a rate of $p=0.5$ is added prior to the final fully connected classifier. This prevents co-adaptation of features, forcing the network to learn redundant and robust representation paths.
- **Target Resolution:** Input image resolution is increased from 64×64 to 128×128 pixels to preserve fine lesion margins, texture details, and pigment

networks.

- **Class-Weighted Cross-Entropy Loss:** To counteract majority-class bias, we introduce inverse-frequency class weights W_c into the loss formulation:

$$L_{\text{Weighted-CE}} = -\frac{1}{N} \sum_{i=1}^N W_{y_i} \log \frac{e^{z_{i,y_i}}}{\sum_j e^{z_{i,j}}}$$

where $W_c \propto \frac{1}{N_c}$, ensuring that misclassifying rare, malignant lesions (e.g., MEL, BCC) incurs a significantly higher gradient penalty than misclassifying NV samples.

3.2 Transfer Learning Model—ResNet18

Transfer learning has proven highly effective for skin cancer classification tasks where the available dataset is limited [2,3]. A recent study found that transfer learning models can significantly enhance the accuracy and efficiency of skin cancer diagnosis, with a transfer-learning-based architecture achieving 97.59% accuracy on the HAM10000 dataset [3]. Building on this success, we employ a ResNet18 model pre-trained on the ImageNet dataset as the starting point. ResNet18 has been extensively validated on HAM10000: comparative studies show it achieves competitive performance ($\approx 80\%$ accuracy) on the seven-class task [4], serves as the backbone of a self-attention architecture that reached 97.5% classification accuracy [5], and is among the effective lightweight CNNs for melanoma detection [6]. Additionally, compared to ResNet50, ResNet18 is more computationally efficient, with only 11 million parameters, making it suitable for training on standard CPU hardware while maintaining high representational capacity.

Therefore, we employ an ImageNet-pre-trained ResNet18 and fine-tune it on the HAM10000 dataset. We adopt a full fine-tuning strategy: all layers of the pre-trained network are trainable throughout training, and the final fully connected layer is replaced with a new linear layer outputting 7 logits corresponding to the seven diagnostic categories (MEL, NV, BCC, AKIEC, BKL, DF, VASC). The entire network is then fine-tuned end-to-end on the HAM10000 training data, allowing the model to adapt its features specifically to the unique visual characteristics of

dermoscopic skin lesion images.

We used a batch size of 32, trained for 15 epochs, and employed the Adam optimizer with an initial learning rate of 1e-4. A ReduceLROnPlateau scheduler with factor 0.5 and patience 5 was applied. The input images were resized to 224×224. Table 5 summarizes the key hyperparameters used in our experiments.

Table 5. Hyperparameters of ResNet18

Hyperparameters	Value
BATCH_SIZE	32
EPOCHS	15
LEARNING_RATE	1e-4
IMG_SIZE	224 pixels
Optimizer	Adam
Loss Function	CrossEntropyLoss
Scheduler	ReducLROnPlateau, mode='min', factor=0.5, patience=5

3.2.1 Forward Propagation and Residual Learning

The core innovation of ResNet18 is the residual block, which addresses the degradation problem in very deep networks. Instead of directly learning a desired underlying mapping $H(x)$, the network learns a residual mapping $F(x) = H(x) - x$. The output of a residual block is defined as:

$$y = F(x, \{W_i\}) + x$$

where x is the input to the block, $F(x, \{W_i\})$ represents the residual function (typically two convolutional layers with batch normalization and ReLU), and y is the block output. The identity shortcut connection $+x$ adds the input directly to the output of the stacked layers. This formulation ensures that if the residual function

becomes zero, the block still performs an identity mapping, which prevents performance degradation as depth increases.

The entire ResNet18 architecture follows a structured forward pass:

- **Initial convolution and pooling:** The input image $X \in R^{224 \times 224 \times 3}$ passes through a 7×7 convolution layer (64 filters, stride 2), batch normalization, ReLU and a 3×3 max-pooling layer (stride 2), producing a feature map of size $56 \times 56 \times 64$.
- **Four residual stages:** The network contains four stages, each with two residual blocks. The stage transitions double the number of channels while halving the spatial dimensions via shortcut down-sampling (stride-2 convolutions). The progression is:

Table 6. Stage of ResNet18

Stage	Input size	Output size	Channels
1	56×56	56×56	$64 \rightarrow 64$
2	56×56	28×28	$64 \rightarrow 128$
3	28×28	14×14	$128 \rightarrow 256$
4	14×14	7×7	$256 \rightarrow 512$

- **Global average pooling (GAP):** Instead of flattening, ResNet18 applies GAP, which computes the average of each 7×7 feature map, resulting in a $1 \times 1 \times 512$ vector. This reduces the number of parameters and mitigates overfitting.
- **Fully connected layer:** The 512-dimensional vector is passed to a linear layer that outputs logits $z \in R^7$ for the seven skin lesion classes.

Finally, the logits are converted to a probability distribution using the softmax function:

$$p_k = \frac{e^{z_k}}{\sum_{j=1}^7 e^{z_j}}, k = 1, \dots, 7$$

Where p_k is the predicted probability that the input belongs to class k .

3.2.2 Loss Function: Cross Entropy with Class Weights

For a multi-class classification task, the standard loss function is categorical cross-entropy. Given the true one-hot encoded label y_{true} and the predicted probabilities p_c for class c , the loss for a single sample is:

$$L = - \sum_{c=1}^7 y_{true,c} \cdot \log(p_c)$$

Because y_{true} has a value of 1 only for the correct class c^* , this simplifies to $L = -\log(p_{c^*})$. The loss approaches 0 as the predicted probability of the correct class approaches 1.

To address class imbalance, we incorporate class weights w_c into the loss function:

$$L = - \sum_{c=1}^7 w_c \cdot y_{true,c} \cdot \log(p_c)$$

The weights are computed based on the inverse of class frequencies, then normalized so that their sum equals the number of classes. Specifically, let N_c be the number of training samples in class c and $K = 7$ the total number of classes. The weight for class c is given by:

$$w_c = \frac{K}{\sum_{j=1}^K \frac{1}{N_j}} \cdot \frac{1}{N_c}$$

This formulation ensures that minority classes receive higher weights while preserving the scale of the loss.

3.2.3 Backpropagation and Gradient Descent

Training a deep neural network requires minimizing the loss function L with respect to the network parameters Θ (all weights and biases). This is achieved through

backpropagation combined with a gradient-based optimization algorithm.

The backpropagation algorithm computes the gradient of L with respect to each parameter using the chain rule of calculus. For a parameter θ at any layer, the gradient is:

$$\frac{dL}{d\theta} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{d \text{intermediate}} \dots \frac{d \text{intermediate}}{d\theta}$$

where $\hat{y}$ represents the model's output before softmax or the loss calculation. Gradients flow from the output layer back to the input layer, allowing all parameters to be updated.

Once gradients are computed, an optimizer updates the parameters by taking a step in the direction opposite to the gradient (gradient descent). The basic update rule is:

$$\theta_{new} = \theta_{old} - \eta \cdot \frac{dL}{d\theta_{old}}$$

where η is the learning rate. In this project, we use the **Adam optimizer**, which extends vanilla gradient descent by maintaining adaptive learning rates for each parameter using estimates of first and second moments of the gradients. The Adam update rule can be summarized as:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \\ \theta_{t+1} &= \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \end{aligned}$$

where $g_t = \frac{dL}{d\theta_t}$ is the gradient at timestep t , and $\beta_1, \beta_2, \epsilon$ are hyperparameters (set to default values 0.9, 0.999, and 10^{-8} respectively). Adam combines the benefits of momentum (to accelerate convergence) and adaptive learning rates (to handle sparse gradients), making it particularly suitable for training deep CNNs on imbalanced medical image data.

3.2.4 Optimizer and Training Duration

The Adam (Adaptive Moment Estimation) optimizer was selected for training the ResNet18 model with an initial learning rate of 1×10^{-4} . Adam is widely regarded as the most effective optimization method for training deep neural networks due to its computational efficiency, ease of use, and strong performance when handling large-scale datasets and high-dimensional parameter spaces [2]. Its adaptive learning rate mechanism makes it particularly well-suited for medical image classification tasks such as skin lesion recognition on the HAM10000 dataset, a setting in which Adam has been consistently adopted in recent literature [7]. Compared to standard SGD, Adam requires less manual tuning and converges more reliably when fine-tuning pretrained networks.

The model was trained for 15 epochs, which was considered sufficient given that ResNet18 was initialized with ImageNet pretrained weights. Transfer learning significantly reduces the number of training epochs required to reach convergence, as the feature extractor is already well-initialized for visual tasks. Additionally, a ReduceLROnPlateau scheduler was employed to halve the learning rate if validation loss plateaued for five consecutive epochs, providing further stability within this training budget.

4. Experimental results and analysis

4.1 Evaluation Metrics

To evaluate the performance of the skin lesion classification models, we adopted multiple metrics, including Accuracy, Precision, Recall, F1-score, Macro-F1, and the Confusion Matrix.

Accuracy measures the overall proportion of correctly classified images. However, because the HAM10000 dataset is highly imbalanced (NV:60%others:40%), accuracy

alone may be misleading.

Precision evaluates how many samples predicted as a certain class are actually correct.

Recall measures the ability of the model to identify all samples belonging to a class.

F1-score is the harmonic mean of precision and recall.

Macro-F1 is particularly important for this study because it assigns equal weight to each class regardless of class frequency, providing a more balanced evaluation under class imbalance.

In addition, confusion matrices were analyzed to identify common misclassification patterns among different skin lesion categories.

4.2 CNN Model Result

We trained both networks for 15 epochs using the Adam optimizer with a learning rate of $\eta=10^{-3}$ and a batch size of 32.

4.2.1 Quantitative Performance Summary

The quantitative results on the validation partition highlight the critical trade-off between global accuracy and class-specific sensitivity under severe imbalance:

Table 7. CNN Performances Summary				
Model Architecture	Accuracy	Macro Precision	Macro Recall	Macro F1-Score
Baseline Custom CNN (Epoch 15)	74.97%	0.5901	0.5118	0.5337
Advanced CNN (Epoch 13)	59.59%	0.4571	0.6001	0.4758

The Baseline CNN achieves a robust global accuracy of 74.97%. However, when examined through the lens of class imbalance, this metric alone does not fully reflect the model's performance on minority categories. While the Baseline achieves a strong Macro-F1 of 0.5337, its performance is still heavily tied to the majority class distribution.

In contrast, the Advanced CNN adopts a different learning strategy that prioritizes

sensitivity. By sacrificing some global accuracy (dropping to 59.59%), the model demonstrates a more aggressive approach to detecting rare lesions. Crucially, its **Macro-Recall increases to 0.6001** (compared to the Baseline's 0.5118), indicating a significantly superior capability to identify critical, under-represented classes that are often overlooked by standard architectures. This reflects a deliberate trade-off where the model trades raw accuracy for higher sensitivity across the entire diagnostic spectrum.

4.2.2 Deep Interpretation of Training Dynamics (Learning Curves)

4.2.2.1 Analysis of Baseline CNN Learning Curves

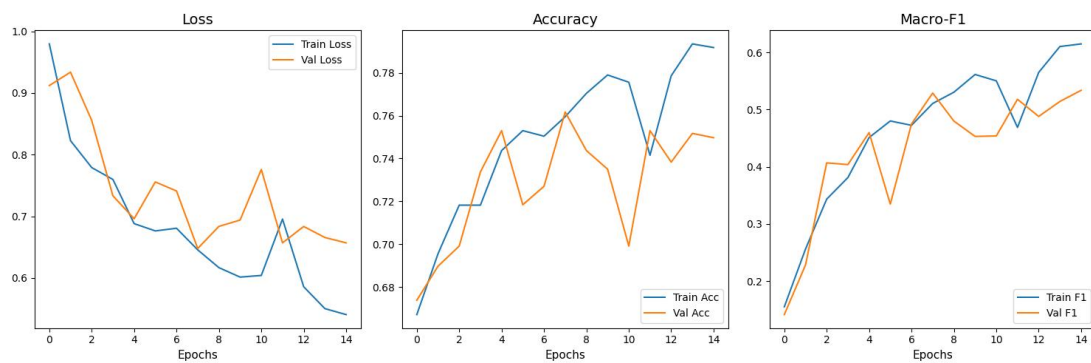


Figure 3. Training and validation learning curves (Loss, Accuracy, and Macro-F1) of the unregularized Baseline CNN

Observation and Trend:

As shown in the training curves (Figure 6.2.1), the Baseline CNN exhibits typical signs of moderate overfitting and optimization instability. During the 15-epoch training run, the Training Loss decreases steadily from 0.9795 to 0.5406, and the Training Accuracy rises from 66.71% to 79.18%. The Validation Loss does not diverge immediately; it decreases from 0.9120 to a minimum of **0.6478** at Epoch 8, before fluctuating significantly (peaking at **0.7759** at Epoch 11) and ultimately settling at **0.6569** by Epoch 15. The Validation Accuracy increases from **67.38%** in the first epoch, peaks at 76.17% at Epoch 8, and fluctuates to finish at **74.97%** at Epoch 15.

Theoretical Interpretation:

The training curves reveal an initial phase of "majority-class bias," where high accuracy masked poor discriminatory performance (reflected by a very low initial F1-score). While the model progressively learned features from the minority classes, the subsequent divergence and instability between training and validation metrics signify clear overfitting. These results demonstrate that a basic CNN lacks the robustness required for this imbalanced dataset.

4.2.2.2 Analysis of Advanced CNN Learning Curves

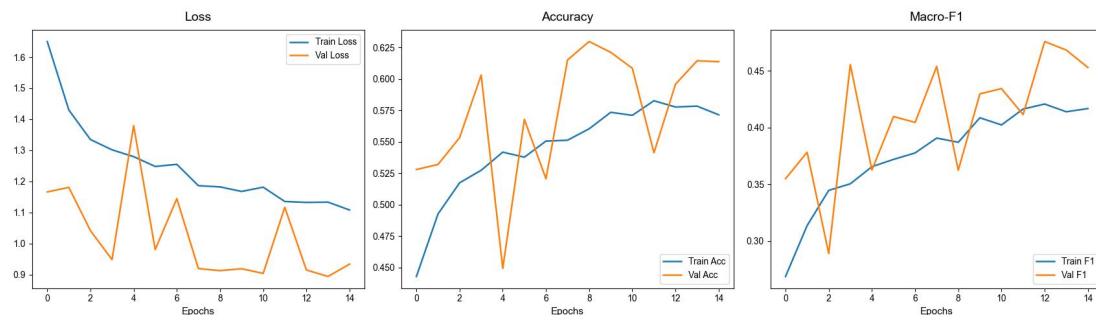


Figure 4. Training and validation learning curves (Loss, Accuracy, and Macro-F1) of the Advanced CNN

Observation and Trend:

The learning curves for the Advanced CNN demonstrate a steady, consistent decline in Training Loss, indicating that the model is effectively learning the representation of the training data. However, the Validation metrics (Loss, Accuracy, and Macro-F1) exhibit significant volatility across the 15 epochs, contrasting with the smooth training trend.

Validation accuracy oscillates noticeably, starting at 52.80% in Epoch 1, peaking at 62.98% in Epoch 9, and fluctuating thereafter to end at 61.38% (with a final test accuracy of 56.19%). Notably, there are visible performance shocks, such as the spike in validation loss in Epoch 5 (1.3793) and a secondary rebound in Epoch 12 (1.1163). These fluctuations suggest that while the training process is stable, the validation performance is highly sensitive to the specific batch composition in each epoch, typical of small or heterogeneous validation sets.

Theoretical Interpretation:

The observed volatility suggests that while architectural components like Dropout and Batch Normalization are functioning to regulate the network and prevent massive overfitting, they are not sufficient to eliminate sensitivity to the dataset's high intra-class variance.

The discrepancy between the smooth training curve and the erratic validation curve indicates that the model is struggling to generalize uniformly across different lesion categories. The periodic spikes in validation loss often coincide with difficulties in correctly classifying samples that share visual similarities. Furthermore, the class-weighted loss, while intended to mitigate the effects of an imbalanced dataset, may be causing these fluctuations if the model's weight updates are frequently reacting to the minority classes within the validation splits. The model achieves moderate success but remains sensitive to the inherent complexity and potential label noise within the dermoscopic dataset.

4.3 ResNet18 Model Result

4.3.1 Experiment A – Baseline ResNet18

Experiment A used a fully fine-tuned ResNet18 model without data augmentation and without class weighting.

a) Training and Validation Result

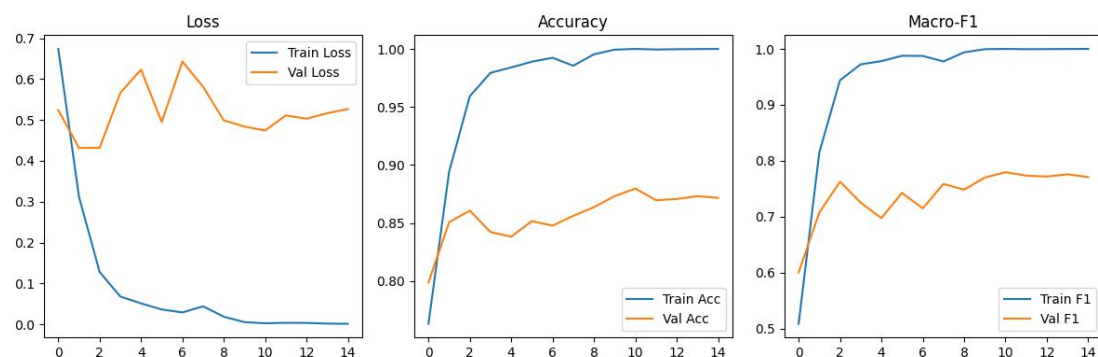


Figure 5. Training and validation learning curves of the Baseline ResNet18

- **Loss:** Training loss dropped rapidly from 0.67 to 0.0016; validation loss started at

0.524, decreased to 0.4315 at epoch 2, then fluctuated and slightly increased to ~0.48–0.53.

- **Accuracy:** Training accuracy reached 100% at epoch 11; validation accuracy peaked at 87.97% (epoch 11) and then stayed around 87%.

- **Macro-F1:** Training macro-F1 reached 1.0; validation macro-F1 peaked at 0.7795 (epoch 11).

Although the model achieved excellent training performance (training accuracy approaching 100%), a noticeable gap existed between training and validation metrics. This indicates that the model started to overfit after several epochs. The validation performance peaked around Epoch 11 and then remained relatively stable while training performance continued improving.

b) Test set Result

On the independent test set (n=1511), the model obtained:

- **Accuracy** = 79.62%
- **Macro-F1** = 0.6630
- **Avg Macro Precision** = 0.7257
- **Avg Macro Recall** = 0.6273

Table 8 .Test results on each classes

	precision	recall	f1-score	support
NV	0.8603	0.9427	0.8996	908
MEL	0.6757	0.4386	0.5319	171
BKL	0.6500	0.6590	0.6545	217
BCC	0.7468	0.6344	0.6860	93
AKIEC	0.5094	0.6279	0.5625	43
VASC	0.8636	0.5429	0.6667	35
DF	0.7742	0.5455	0.6400	44

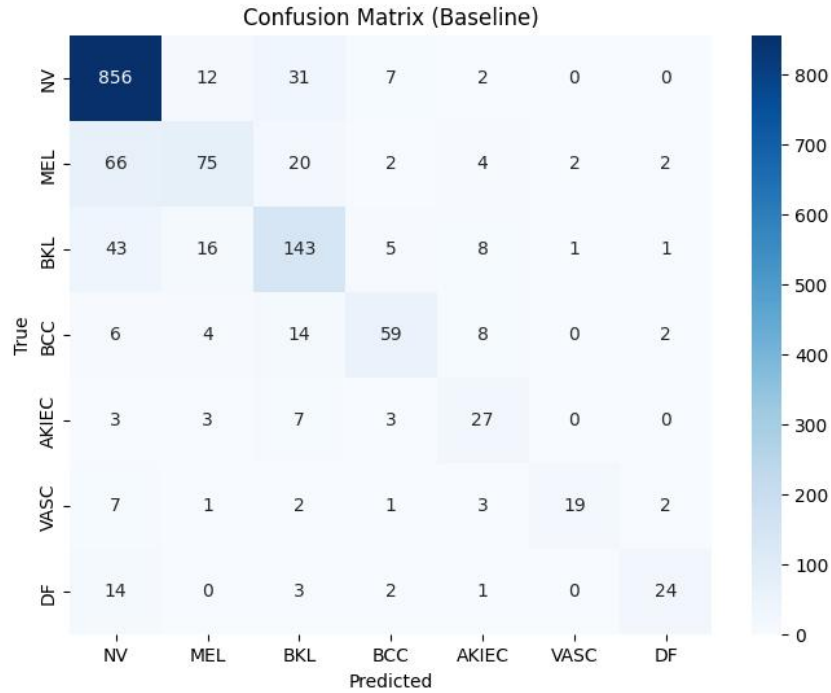


Figure 6. Confusion Matrix of Baseline Resnet18

The confusion matrix shows that the model classified the majority class (NV) very well, achieving an F1-score of 0.8996, while minority classes such as MEL (F1=0.53), AKIEC (F1=0.56), and DF (F1=0.64) had significantly lower scores, indicating a typical symptom of class imbalance.

Therefore, Experiment A serves as a strong baseline but also reveals class imbalance issue, potential overfitting and limited generalization to unseen data.

4.3.2 Experiment B – ResNet18 with Data Augmentation

To address the overfitting issue observed in Experiment A, Experiment B introduced data augmentation during training, including random horizontal flipping, vertical flipping, and random rotation. No class weighting was applied.

a) Training and Validation Result

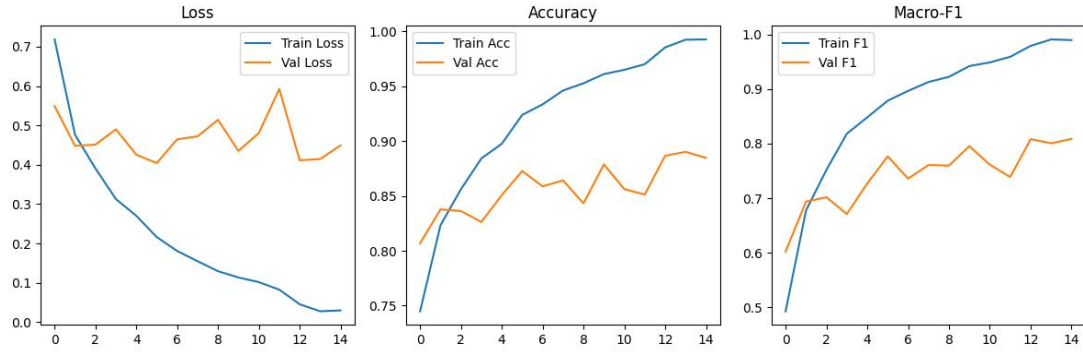


Figure 7. Training and validation learning curves of the ResNet18 (with Data Augmentation)

- **Loss:** Training loss decreased from 0.718 to 0.03; validation loss fluctuated between 0.40 and 0.59, with a minimum of 0.4042 at epoch 6.
- **Accuracy:** Training accuracy reached 99.3%; validation accuracy peaked at 89.02% (epoch 14).
- **Macro-F1:** Training macro-F1 up to 0.991; validation macro-F1 peaked at 0.8083 (epoch 13).

The accuracy curves suggest that gap between train acc and val acc has narrowed , which means data augmentation reduced overfitting problem compared to Exp-A by generating more diverse training samples. As a result, the model learned more robust visual features and became less dependent on specific image orientations or appearances.

b. Test set Result

On the test set, the model achieved:

- **Accuracy** = 78.95%
- **Macro-F1** = 0.6670
- **Avg Macro Precision** = 0.7333
- **Avg Macro Recall** = 0.6221

Table 9 .Test results on each classes

	precision	recall	f1-score	support
NV	0.8311	0.9482	0.8858	908
MEL	0.5753	0.4912	0.5300	171
BKL	0.8082	0.5438	0.6501	217
BCC	0.7059	0.6452	0.6742	93
AKIEC	0.5714	0.5581	0.5647	43
VASC	0.8077	0.6000	0.6885	35
DF	0.8333	0.5682	0.6757	44

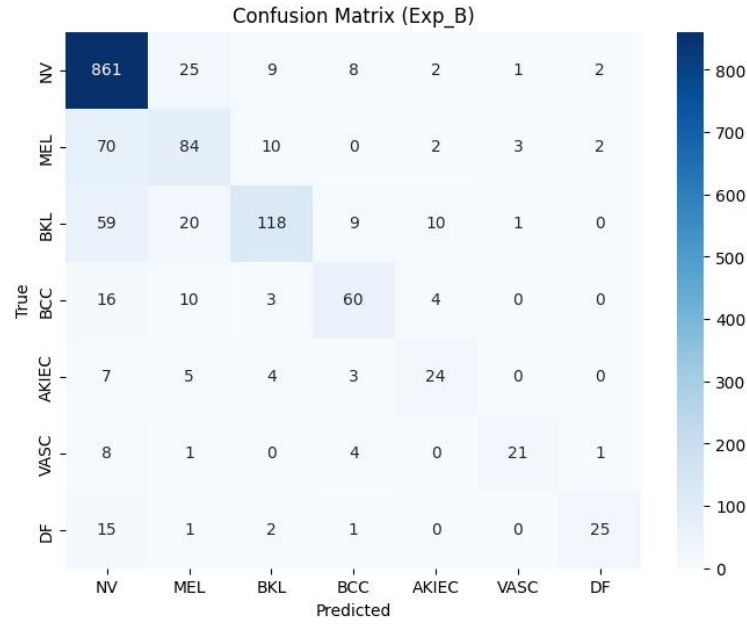


Figure 8. Confusion Matrix of Resnet18 (with Data Augmentation)

Compared with the baseline model, data augmentation slightly improved Macro-F1 (0.6670 vs. 0.6630) and Macro Precision (0.7333 vs. 0.7257), indicating better recognition of minority classes. However, the overall test accuracy decreased slightly from 79.62% to 78.95%.

Although the accuracy improvement was limited, the increase in Macro-F1 demonstrates that data augmentation enhanced class-balanced performance and improved the model's generalization ability.

4.3.3 Experiment C – ResNet18 with Data Augmentation & Class Weighting

Experiment C combined both data augmentation and class weighting during training. Data augmentation included random horizontal and vertical flips and random rotations, while class weights were computed to counter the severe class imbalance in the HAM10000 dataset.

a) Training and Validation Result

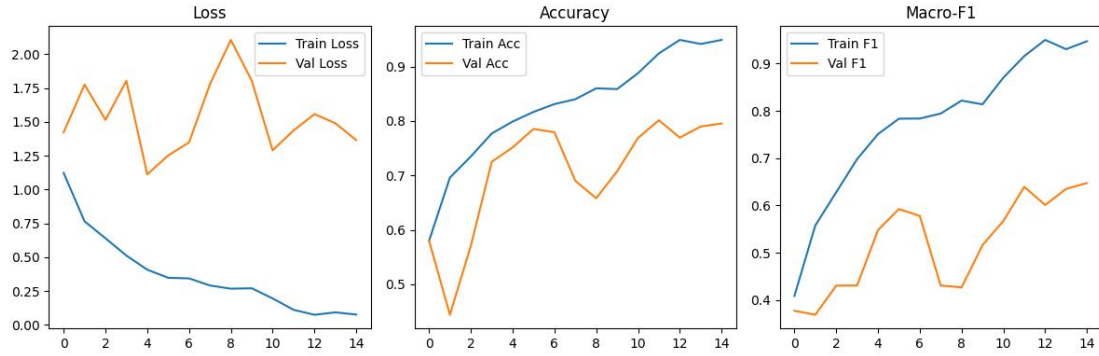


Figure 9. Training and validation learning curves of the ResNet18 (with Data Augmentation and class weighting)

- **Loss:** Training loss decreased steadily from 1.12 to 0.076; validation loss was highly unstable (ranging 1.1–2.1), never below 1.1.
- **Accuracy:** Training accuracy reached 94.9%; validation accuracy peaked at 80.13% (epoch 12) but mostly around 75–78%.
- **Macro-F1:** Training macro-F1 reached 0.95; validation macro-F1 peaked at 0.639 (epoch 12) but was generally low (~0.43–0.55).

b) Test set Result

On the independent test set, the model achieved:

- **Accuracy** = 73.99%
- **Macro-F1** = 0.5815
- **Avg Macro Precision** = 0.6418
- **Avg Macro Recall** = 0.5813

Table 10 .Test results on each classes

	precision	recall	f1-score	support
NV	0.8161	0.9141	0.8623	908
MEL	0.5652	0.1520	0.2396	171
BKL	0.5437	0.6313	0.5842	217
BCC	0.6750	0.5806	0.6243	93
AKIEC	0.6250	0.3488	0.4478	43
VASC	0.5179	0.8286	0.6374	35
DF	0.7500	0.6136	0.6750	44

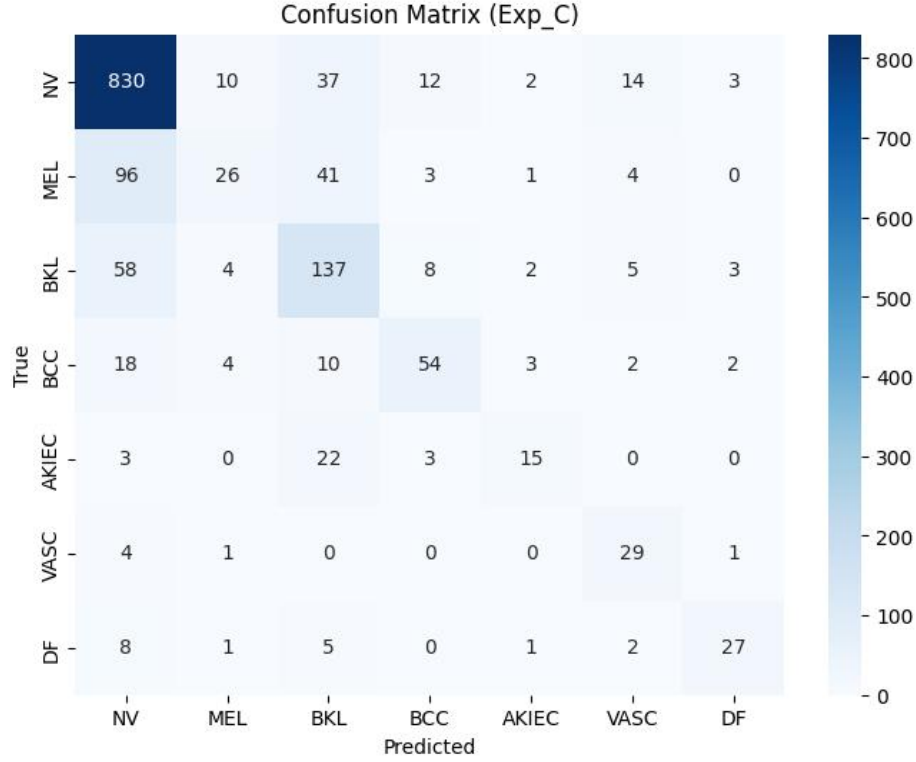


Figure 10. Confusion Matrix of Resnet18 (with Data Augmentation&Class Weighting)

The confusion matrix and per-class F1-scores indicate that the model still performs very well on the majority class (NV, $F1 = 0.8623$), while minority classes benefit from class weighting (e.g., VASC $F1 = 0.6374$, DF $F1 = 0.6750$). However, some classes like MEL still remain challenging ($F1 = 0.2396$), reflecting inherent difficulty in classifying these lesions.

Experiment C, which combined data augmentation and class weighting, achieved lower test accuracy (73.99%) and lower Macro-F1 (0.5815) than Experiments A and B. This suggests that the applied class weighting and data augmentation, in combination, did not improve overall generalization and may have adversely affected the model's performance on the majority classes.

4.4 Comparison of Different Models and Training Strategies

4.4.1. Training Performance Comparison

The training performance reflects each model's capacity to fit the training data and learn discriminative features.

Table 11 .Training results comparison

Model	Train Accuracy	Train Macro-F1	Final Train Loss
Baseline CNN	~79.18%	~0.60 (approx)	0.5406
Advanced CNN	64.00–66.00%	~0.48–0.50	~1.00+
ResNet18-A	75~100%	1.000	0.67~0.0016
ResNet18-B	80~99.3%	0.991	0.718~0.03
ResNet18-C	~94.90%	0.9469	0.0761

Key Comparison:

- The **Advanced CNN shows underfitting/optimization difficulty**, with significantly lower training accuracy compared to all other models.
- All **ResNet18 models achieve stronger training performance**, indicating better feature extraction capability.
- ResNet18-A achieves extremely high training accuracy, suggesting **very strong fitting ability but potential over-optimization**.
- CNN models (especially Advanced CNN) are clearly **capacity-limited compared to ResNet18**, failing to fully exploit training data.

4.4.2. Validation Performance Comparison

Table 12 .Validation results comparison

Model	Best Val Accuracy	Best Val Macro-F1	Stability (Epoch behavior)
Baseline CNN	~74.97%	~0.53	Moderate stability
Advanced CNN	~62.98%	~0.48	High fluctuation
ResNet18-A	87.97%	~0.78	Stable
ResNet18-B	89.02%	~0.81	Stable
ResNet18-C	80.13%	~0.64	Unstable

Key Comparison:

- ResNet18-A and B achieve the best validation performance, both in accuracy and Macro-F1.
- CNN models show weaker validation performance, especially the Advanced CNN, which exhibits large instability across epochs.
- ResNet18-C shows strong training performance but degraded validation performance, indicating overfitting caused by aggressive training strategy

(augmentation + class weighting).

- Overall, ResNet18 models demonstrate better generalization consistency than CNN models.

4.4.3. Test Performance Comparison (Final Generalization Ability)

Table 13 .Test results Comparison

Model	Test Accuracy	Macro-Precision	Macro-Recall	Macro-F1
Baseline CNN	74.97%	0.5901	0.5118	0.5337
Advanced CNN	56.59%	0.4571	0.6001	0.4758
ResNet18-A	79.62%	0.7257	0.6273	0.663
ResNet18-B	78.95%	0.7333	0.6221	0.667
ResNet18-C	73.99%	0.6418	0.5813	0.5815

Key Comparison:

- **ResNet18-A achieves the best overall test performance**, especially Macro-F1.
- ResNet18-B achieves slightly higher accuracy but slightly lower Macro-F1 → indicates **accuracy–balance trade-off**.
- ResNet18-C shows reduced accuracy and Macro-F1, but relatively balanced recall, suggesting **bias reduction at cost of overall performance**.
- Advanced CNN performs worst on test set, confirming that architectural capacity is insufficient for this task.

4.4.4 Cross-Model Insights (CNN vs ResNet18)

1)Architectural Effect

ResNet18 consistently outperforms CNN across all splits.

This indicates that deep residual learning significantly improves feature representation quality, especially for fine-grained medical images.

2)Training Strategy Sensitivity

CNN performance is relatively limited regardless of training strategy.

ResNet18 is more sensitive to training strategies:

A → best accuracy

B → best balanced performance

C → improved minority recall at the start but weaker overall performance compared to A & B

3)Generalization Behavior

CNN models show early saturation and unstable validation trends.

ResNet18 models show higher ceiling but risk of overfitting when over-regularized or heavily reweighted.

Best generalization is achieved by ResNet18-A/B without excessive loss re-weighting.

4.5 Experimental Issues Discussion

In Model Resnet18 part, the three experiments investigate the impact of data augmentation and class weighting on a fully fine-tuned ResNet18 model for skin lesion classification, and this part we will discuss experimental issues including class imbalance, overfitting, data augmentation, and model generalization based on the Resnet18 model results.

Table 14. summarizes the overall test performance of the three experiments:

Experiment	Data Augmentation	Class Weighting	Test Accuracy	Test Macro-F1
Exp A	No	No	79.62%	0.6630
Exp B	Yes	No	78.95%	0.6670
Exp C	Yes	Yes	73.99%	0.5815

a) Class Imbalance

• **Evidence:** The HAM10000 dataset exhibits a severe class imbalance problem. The NV class accounts for the majority of samples, while classes such as DF, VASC, and

AKIEC contain relatively few examples. This imbalance makes the model naturally biased toward predicting majority classes.

From the confusion matrices of Experiment A and B, it is clear that many minority-class samples (e.g., AKIEC, DF, VASC, and MEL) were misclassified as NV. Particularly, the model often confused NV, MEL, and BKL classes, indicating that the model was biased toward the majority class. The majority class NV dominates (908 test samples), achieving high recall (>0.94), while minority classes such as MEL, VASC and DF show much lower recall (0.44 – 0.54). The macro-F1 is consistently lower than the weighted F1 (Exp-A: macro 0.663 vs weighted 0.787), confirming the effect of imbalance.

- **Effect of class weights (Exp-C)** : Contrary to expectation, weighting did not improve minority performance. Instead, it destabilised training – validation macro-F1 dropped to 0.58, and MEL recall collapsed to 0.152. This suggests that simple inverse frequency weighting may be too aggressive for very rare classes (MEL only 1.7% of training data) when combined with full fine-tuning. A milder weighting or oversampling might be more appropriate.

b) Overfitting

Because all Resnet18 experiments employed full fine-tuning, all parameters of ResNet18 were updated, allowing the network to learn rich, task-specific features. Full fine-tuning improves the model's capacity to memorize training data, increasing the risk of overfitting.

In Experiment A, the gap between training and validation performance was substantial (training Macro-F1 >0.9 , validation Macro-F1 ~ 0.66), indicating overfitting. Experiment B, which incorporated data augmentation, shows a reduced gap between training and validation performance, suggesting that augmentation mitigated overfitting by exposing the model to more diverse examples.

c) Effect of Data Augmentation

Comparing Experiment A and B shows that data augmentation slightly lowered test Accuracy (79.62% $\rightarrow$ 78.95%) but improved Macro-F1 (0.6630 $\rightarrow$ 0.6670). This indicates that augmentation helped the model learn more robust features across classes, rather than overfitting to dominant classes. Augmented training examples (rotations, flips) improve generalization to diverse lesion appearances.

d) Model Generalization

Experiment B demonstrates the best trade-off between Accuracy and class-balanced performance, showing that data augmentation improved generalization across different lesion types. Experiment A achieves slightly higher Accuracy due to NV dominance but performs worse on minority classes. Experiment C illustrates that naively adding class weights can improve minority-class recall but may harm overall generalization if not carefully tuned.

In summary, full fine-tuning enables the model to capture rich features, data augmentation improves robustness and reduces overfitting, and class weighting needs careful calibration to balance minority-class performance with overall predictive accuracy.

5. Model interpretation and error analysis

5.1 CNN: Confusion Matrix Diagnosis and Class-Specific Behavior

To understand the decision-making behavior of both models, we analyze their respective 7-class confusion matrices from the validation partition.

5.1.1 Analysis of the Baseline CNN Confusion Matrix

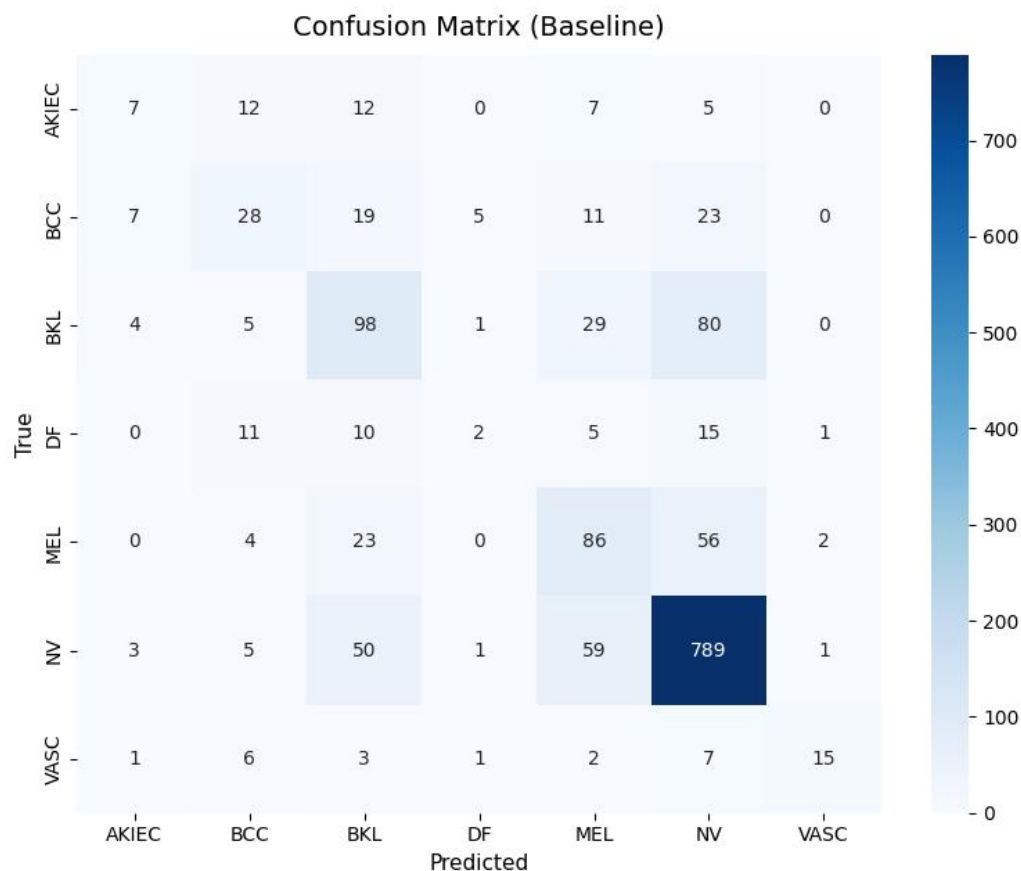


Figure 11. Confusion matrix of the Baseline Custom CNN on the validation partition

Observation and Trend:

The confusion matrix (Figure 7.1.1) reveals a significant predictive bias towards the Melanocytic Nevi (NV) class. While the model correctly identifies instances in other categories—most notably 98 for BKL, 86 for MEL, and 28 for BCC—there is a stark vertical concentration of predictions in the 'NV' column (the 6th column). This indicates that a large portion of samples from other classes (e.g., 80 samples of BKL, 56 samples of MEL) are frequently misclassified as NV. Despite these misclassifications, the diagonal elements show that the model does maintain some discriminative ability, with 789 instances correctly identified as NV, demonstrating the model's heavy reliance on the majority class to achieve its overall accuracy.

Theoretical Interpretation:

This distribution is a direct consequence of using unweighted cross-entropy loss on a

highly imbalanced dataset where NV represents about 60% of the total population (908 out of 1,511 test samples). Because the majority class dominates gradient updates during training, the network shifts its decision boundaries to engulf large portions of the adjacent feature space. This results in severe bias toward predicting NV, producing a high global accuracy (67.84% on the test set) but failing to reliably identify minority and high-risk diseases. To make the model clinically viable, interventions such as weighted loss functions, oversampling, or sample-balanced data augmentation are required.

5.1.2 Analysis of the Advanced CNN Confusion Matrix

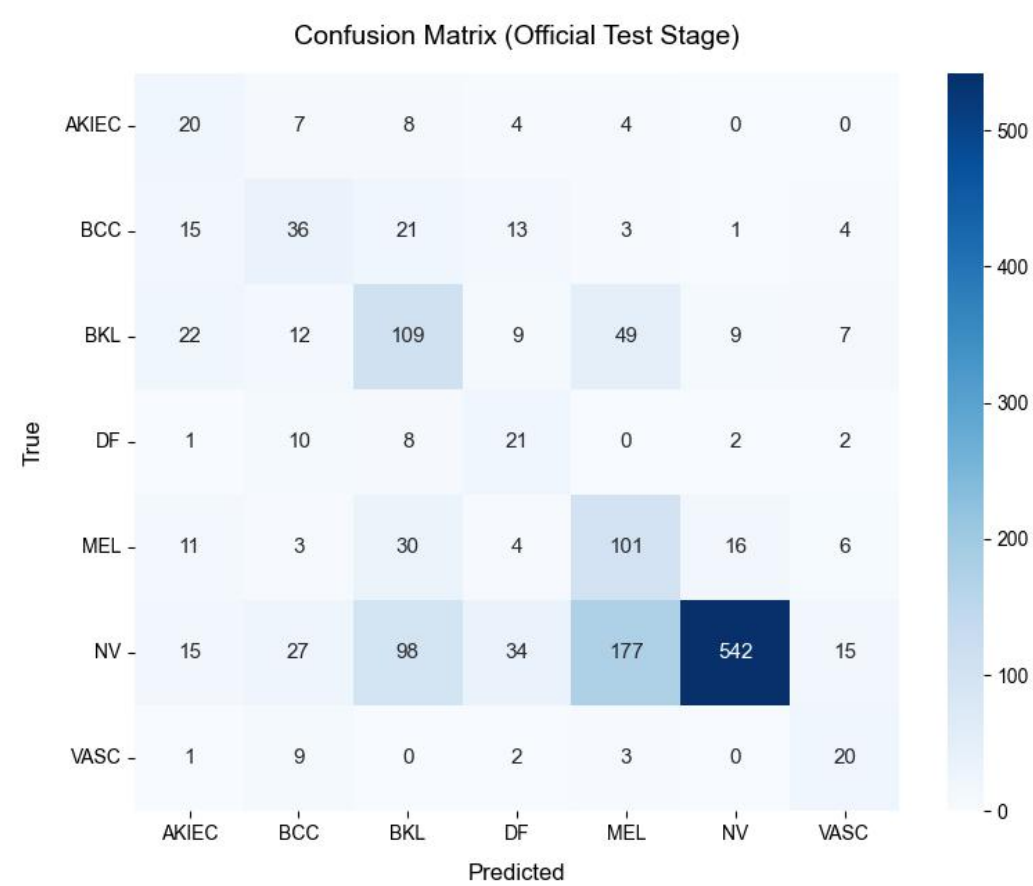


Figure 12. Confusion matrix of the Advanced CNN on the official validation partition

Observation and Trend:

The confusion matrix for the official test set (Figure 7.1.2) reveals significant class imbalance challenges that remain despite the training process. While the model shows

a strong ability to classify the majority class 'NV' (Melanocytic nevus), with 542 correctly identified samples, this success comes at the cost of high classification bias. The model displays a clear "vertical stripe" pattern within the NV column, indicating that when the model is uncertain about samples from minority-class categories like BKL or MEL, it frequently defaults to predicting 'NV'. The total correctly classified counts for the seven categories are as follows:

- AKIEC: 20/43
- BCC: 36/93
- BKL: 109/217
- DF: 21/44
- MEL: 101/171
- NV: 542/908
- VASC: 20/35

While there is some presence along the diagonal for all classes, the model's predictions are not evenly distributed; they are heavily skewed by the majority class, indicating that the model has not yet achieved satisfactory generalization on smaller, rarer diagnostic categories.

Theoretical Interpretation:

The persistence of heavy off-diagonal misclassifications into the 'NV' column suggests that the implemented class-weighted loss function was insufficient to fully decouple the model's decision boundaries from the dataset's inherent class prevalence. The dominant presence of the NV class (the largest category) continues to exert a "gravitational pull" on the model's predictions. The theoretical goal of the inverse-frequency weighting was to penalize minority-class errors; however, the model's inability to prevent the migration of these minority samples into the majority class (NV) implies that the features learned by the CNN, while sufficient for the

majority class, lack the discriminative depth required to distinguish rarer, more complex pathologies from the NV baseline.

5.1.3 Error Analysis and Clinical Case Studies

Despite these overall performance improvements, certain categories remain challenging for the networks to classify, reflecting real-world clinical hurdles in dermatological diagnosis and the visual ambiguities inherent to skin lesions.

5.1.3.1 Case I: Melanoma (MEL) Confused with Melanocytic Nevi (NV)

Early-stage Melanoma (MEL) and benign Melanocytic Nevi (NV) share major overlapping pathophysiological and visual features, such as atypical pigment networks, light hyperpigmentation, and subtle structural variations. This visual similarity leads to significant classification trade-offs between the two models:

- In the Baseline CNN: The model suffers from severe under-sensitivity to malignant cases due to class imbalance. In the validation stage (Epoch 15), 71 out of 167 MEL cases are misclassified as benign NV, indicating a critically high false-negative rate that represents a severe clinical risk.
- In the Advanced CNN: The model successfully mitigates this dangerous clinical loophole, reducing MEL-to-NV misclassifications to only 6 cases in validation and 16 cases in the official test set. However, prioritizing high sensitivity introduces a opposite diagnostic trade-off (false positives): 276 benign NV cases are misclassified as malignant MEL in validation (and 177 cases in the test set).

This diagnostic shift indicates that while the Advanced CNN successfully extracts the high-level features needed to capture subtle melanocytic anomalies, its decision boundary is heavily biased towards over-diagnosing benign nevi as melanomas to minimize critical false negatives.

5.1.3.2 Case II: Benign Keratosis-like Lesions (BKL) Confused with Melanoma (MEL)

As shown in the confusion matrices, Benign Keratosis-like Lesions (BKL) represent another prominent source of false positives, frequently being misclassified as malignant Melanoma (MEL). Benign keratoses often exhibit scaling, hyperpigmentation, and irregular borders that visually mimic the diagnostic hallmarks of melanoma. This overlap presents a significant challenge:

In the Advanced CNN, 38 BKL samples in the validation set (and 49 in the official test set) are incorrectly classified as MEL, making it the largest source of error for this category.

This confusion is exacerbated by clinical artifacts such as hair-follicle blockages, superficial scaling, and variations in dermatoscopic lighting. These external factors introduce high-frequency noise that easily overwhelms the shallow receptive fields of the custom convolutional layers, obscuring the underlying fine tissue textures and leading to misclassifications.

5.2 ResNet18

- **Failed Case (Figure 13.): True: MEL → Pred: NV (Conf: 1.000)**

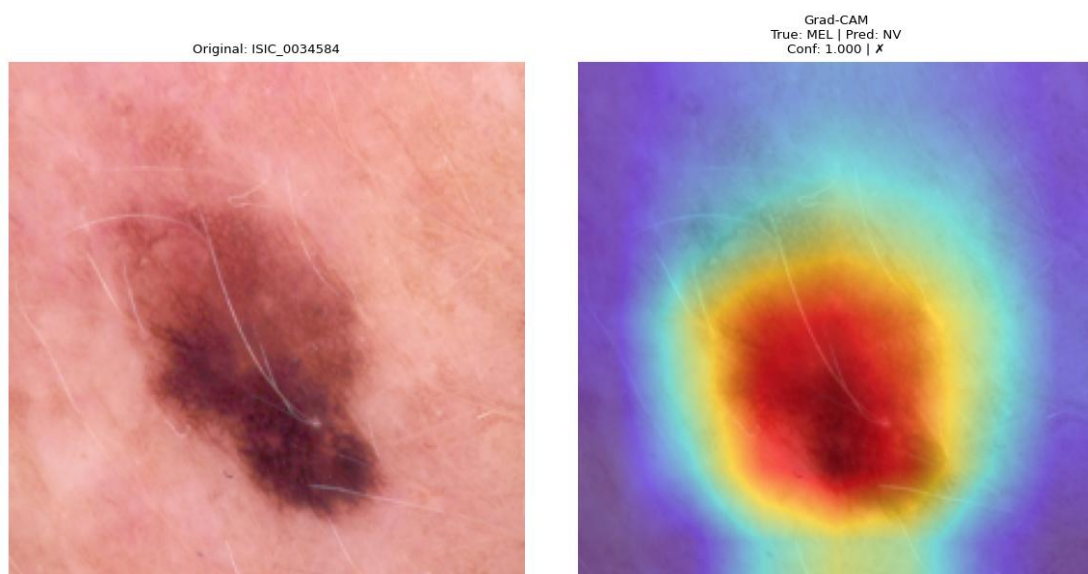


Figure 13. Failed Case of ResNet18

The Grad-CAM heatmap reveals that the model concentrated activation on the central body of the lesion while largely ignoring its peripheral regions. Although this lesion exhibits key melanoma indicators — including asymmetric morphology, irregular borders, and heterogeneous pigmentation (notably the darker region in the inferior portion) — the model assigned NV with 100% confidence. This suggests that ResNet18 failed to capture the diagnostically critical border irregularities and color variation, instead responding to the overall lesion shape in a manner consistent with its learned NV representations. The extremely high confidence in an incorrect prediction further indicates poor model calibration for visually ambiguous cases.

- **Successful Case (Figure 14.): True: NV → Pred: NV (Conf: 1.000)**

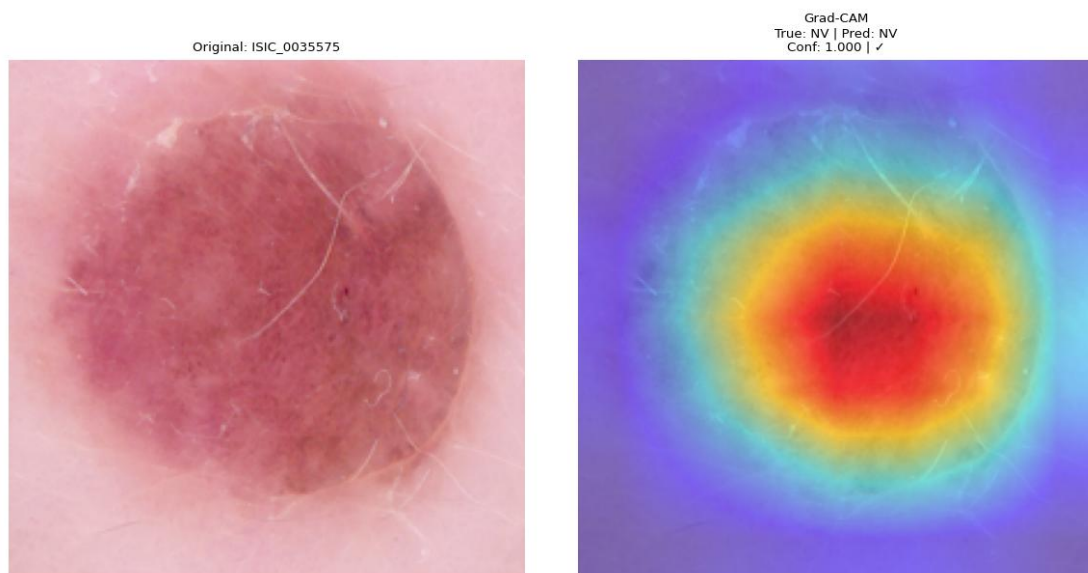


Figure 14. Successful Case of ResNet18

In contrast, the Grad-CAM heatmap for the correctly classified NV lesion shows strong, well-centered activation across the entire lesion body. The lesion presents symmetric, round morphology with uniform pinkish-red coloration — features that strongly align with typical NV characteristics. The model's focused and coherent activation pattern suggests that it successfully identified the relevant discriminative features, resulting in a high-confidence correct prediction.

- **Comparative Insight**

Notably, both cases produced Grad-CAM patterns with centrally concentrated

activations, yet only one prediction was correct. This highlights a key limitation: the model relies predominantly on global lesion shape and color distribution rather than fine-grained local features such as border irregularity or focal pigmentation variation. In the failed case, the MEL lesion's partial visual resemblance to NV — particularly in its upper region — was sufficient to mislead the model, consistent with the broader MEL – NV confusion pattern observed in the confusion matrices.

6. Conclusion

This project investigated deep learning approaches for skin lesion classification using the HAM10000 dermoscopic dataset across seven diagnostic categories. A custom CNN baseline and three variants of a ResNet18 transfer learning model were systematically designed, trained, and evaluated using PyTorch, with results assessed on an independent test set using accuracy, Macro-F1, precision, recall, and confusion matrices.

The experimental results consistently demonstrated that transfer learning substantially outperforms custom CNN architectures in this task. Even the simplest ResNet18 variant achieved markedly higher accuracy and Macro-F1 than the CNN baseline, with the improvement attributable to the rich visual representations learned from large-scale pretraining on ImageNet. Introducing data augmentation into the CNN pipeline yielded only marginal gains, suggesting that augmentation alone is insufficient to compensate for the limited feature extraction capacity of a shallow custom network.

Among the ResNet18 experiments, the augmentation-only variant achieved the best overall generalization performance, slightly surpassing the baseline while exhibiting a narrower training–validation gap, indicating that augmentation effectively reduced overfitting by exposing the model to more diverse visual patterns during training. In contrast, combining data augmentation with class weighting yielded the weakest performance among the ResNet18 variants. Although class weighting improved recall

for certain minority classes such as VASC, it destabilized training and significantly degraded performance on MEL, suggesting that the interaction between aggressive class reweighting and augmentation introduced optimization difficulties rather than consistent improvements.

6.1 Challenges and Limitations

Class imbalance was the most persistent challenge throughout all experiments. The NV class comprises approximately 60% of the dataset, and all models — regardless of architecture or training strategy — exhibited a strong bias toward correctly classifying NV while underperforming on minority classes such as VASC, AKIEC, and DF.

Overfitting was observed in multiple experiments, most notably in Experiment A, where training accuracy reached 100% while validation accuracy stabilized around 87–88%. The large gap between training and validation/test performance highlights that the model memorized training samples rather than learning generalizable features. Data augmentation in Experiments B and C partially mitigated this issue but could not fully eliminate it, partly because the dataset size is still limited relative to the model complexity.

Inter-class visual similarity also contributed to misclassification. In the CNN experiments, the confusion matrices revealed a strong majority-class bias, with minority-class samples frequently misclassified as NV, the dominant class. In contrast, the ResNet18 experiments demonstrated a different confusion pattern, where BKL, MEL, and NV were frequently confused with one another, suggesting that the visual similarity among these lesion types posed a persistent classification challenge even for a pretrained deep learning model.

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